

Term of Reference 1

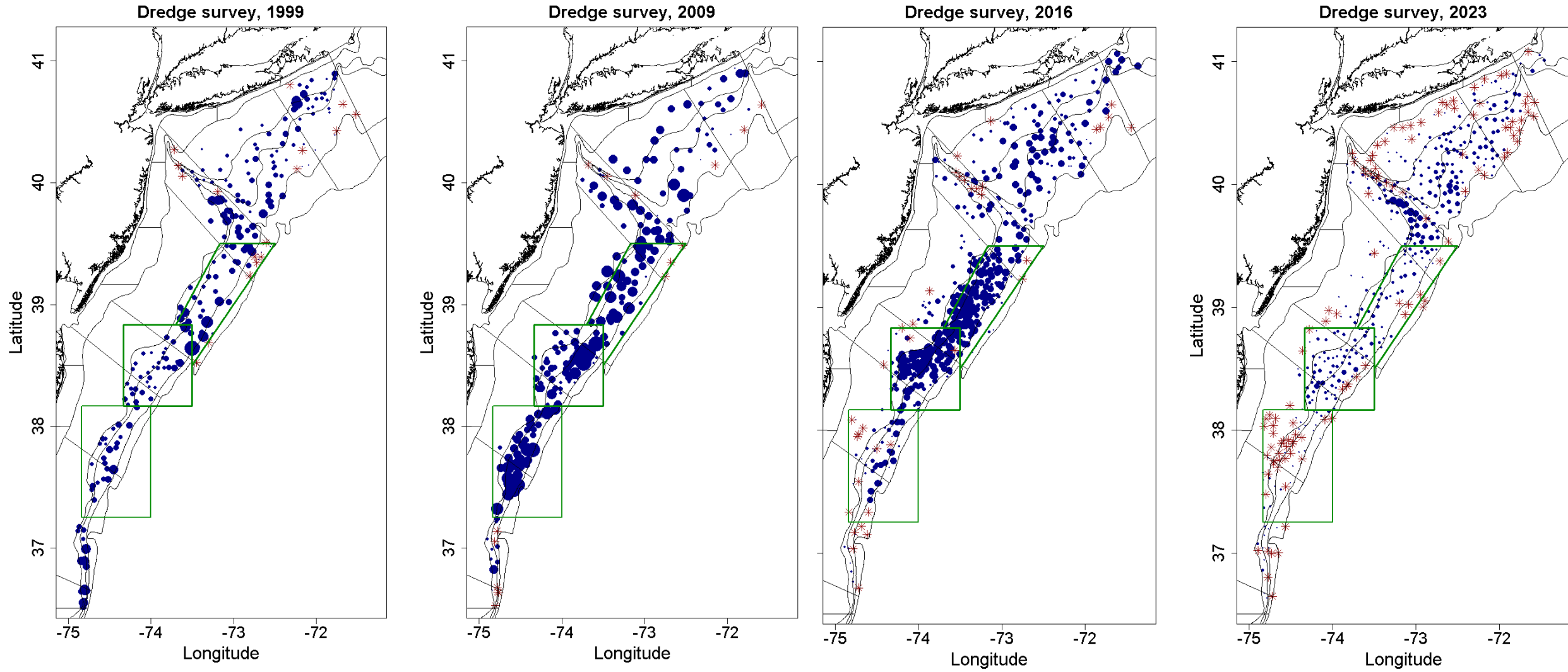
Identify relevant ecosystem and climate influences on the stock. Characterize the uncertainty in the recent sources of data and their link to stock dynamics. Consider findings, as appropriate, in addressing other Terms of Reference. Report how the findings were considered under impacted Terms of Reference.



Scallop biomass from dredge surveys, 1999-2023

Successive truncation of southern and inshore distribution

★ = zero scallops observed



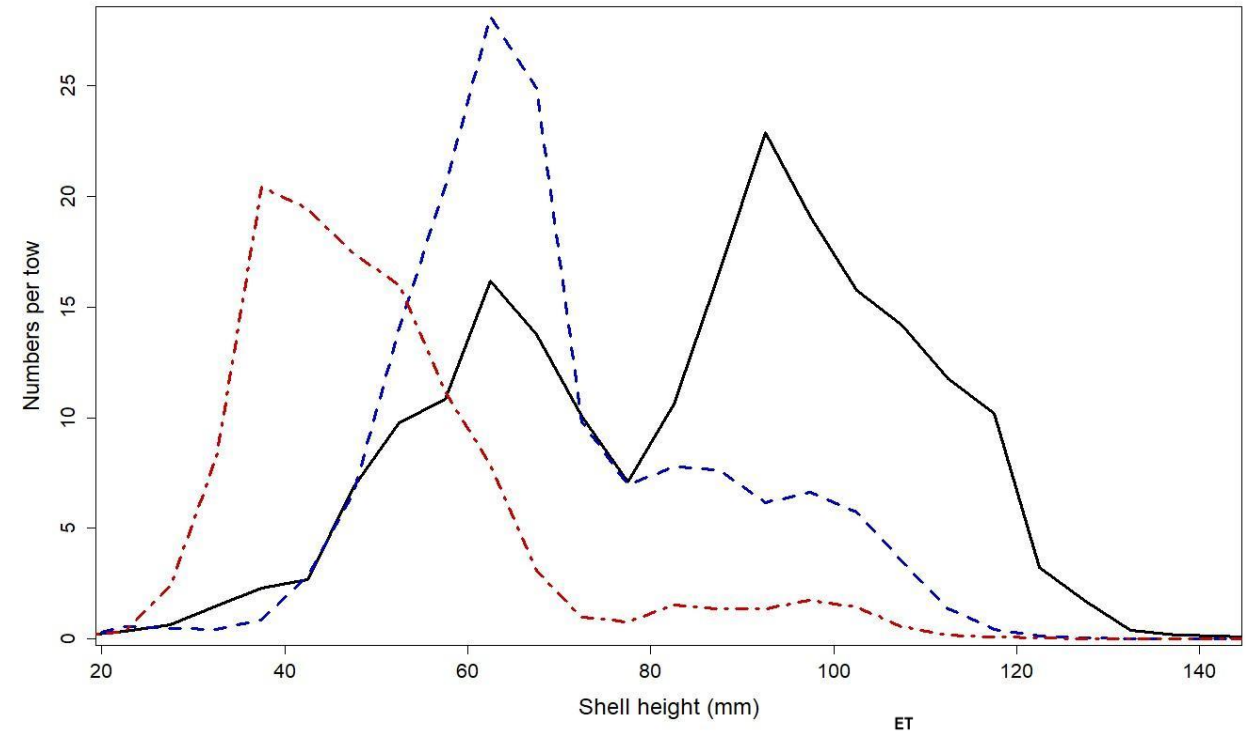
Size-frequencies from the Virginia area in

1998-2000 (Black line)

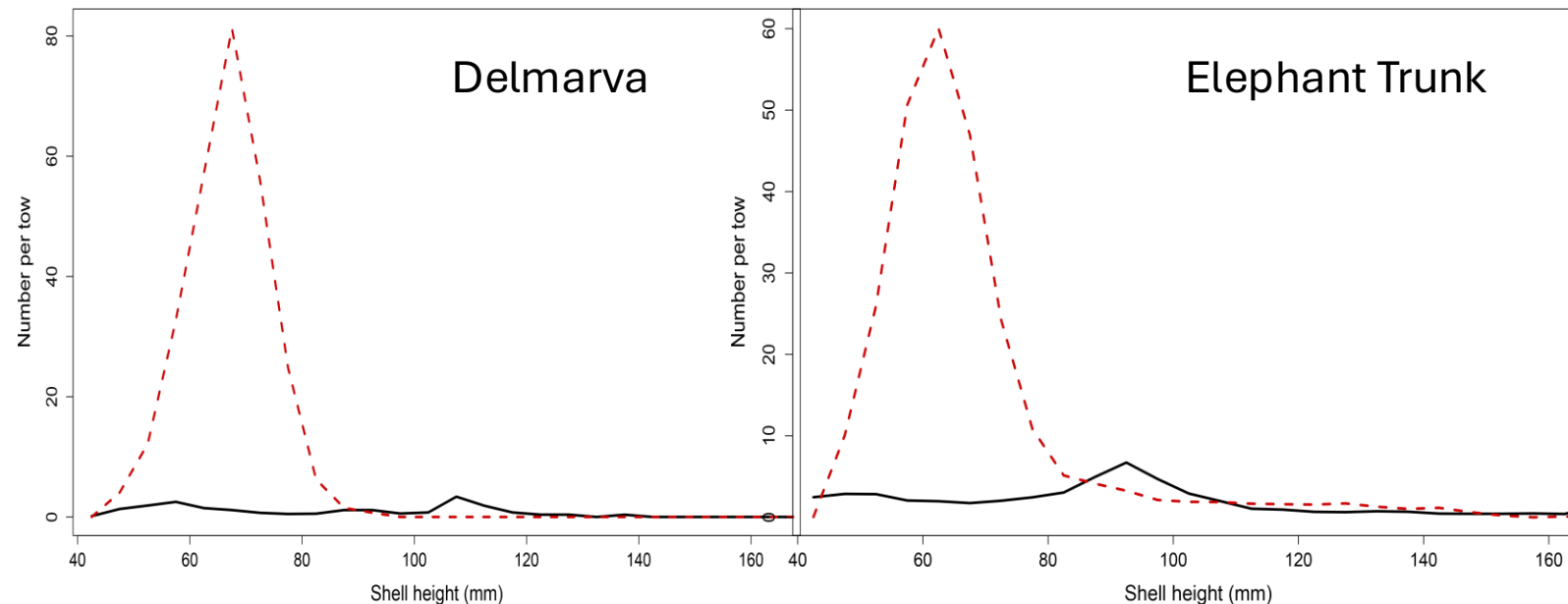
2001-2008 (Blue dashed line)

2009-2018 (Red dashed dotted line)

showing reduced survival of age 3+ scallops



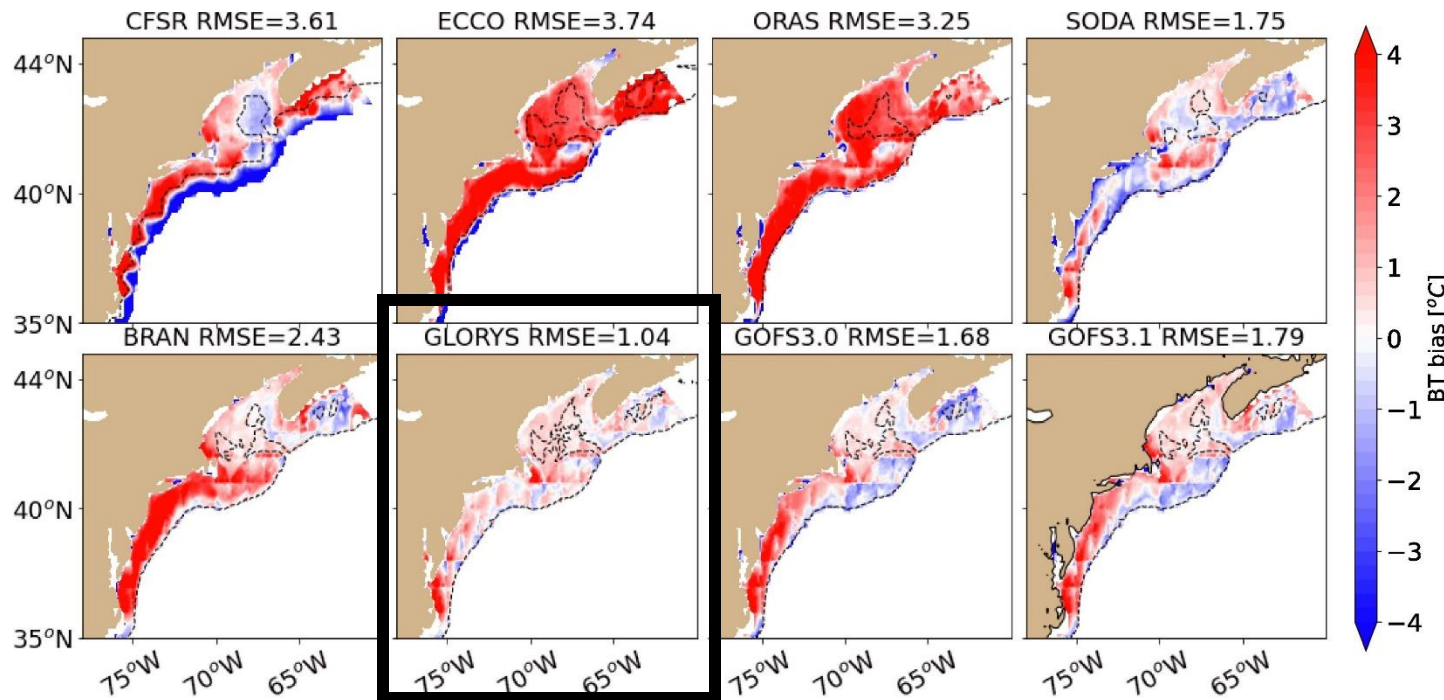
Size-frequencies in Delmarva and Elephant Trunk in 2022 (red) and 2023 (black), showing low survival beyond age 2.



TOR1: Climate & Environmental Considerations

An exploration of thermal habitat changes and risk in the Mid-Atlantic Bight using GLORYS, 1993-2023

Joseph Caracappa, Deborah Hart, Jui-Han Chang, Kimberly Hyde



Objectives

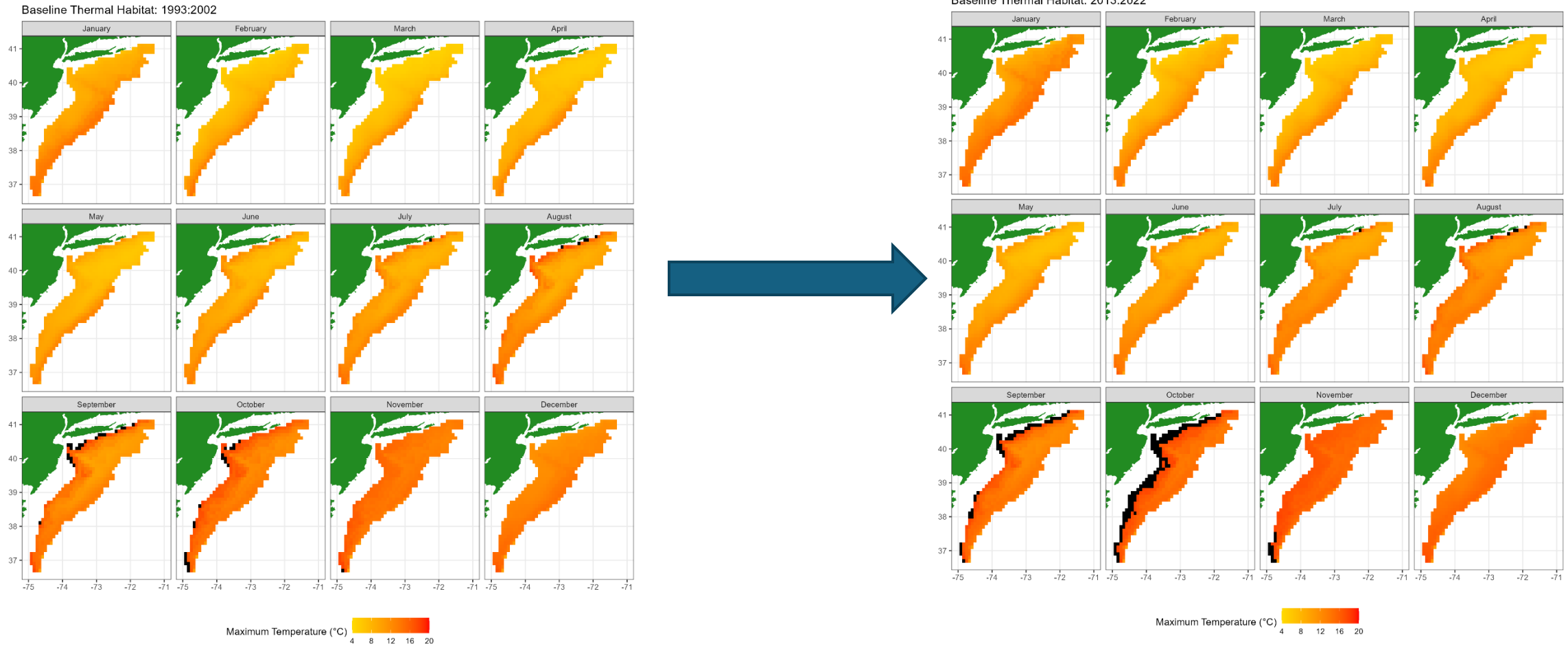
- Characterize thermal habitat for sea scallops
- Develop thermal stress risk indicators
- Identify changes over time

Using GLORYS reanalysis for bottom temperature

- A global product from Copernicus
- 1/12° resolution
- Daily from 1993 – present
- Top performing reanalysis product for bottom temperature in northeast US

Castillo-Trujillo et al.,
2023

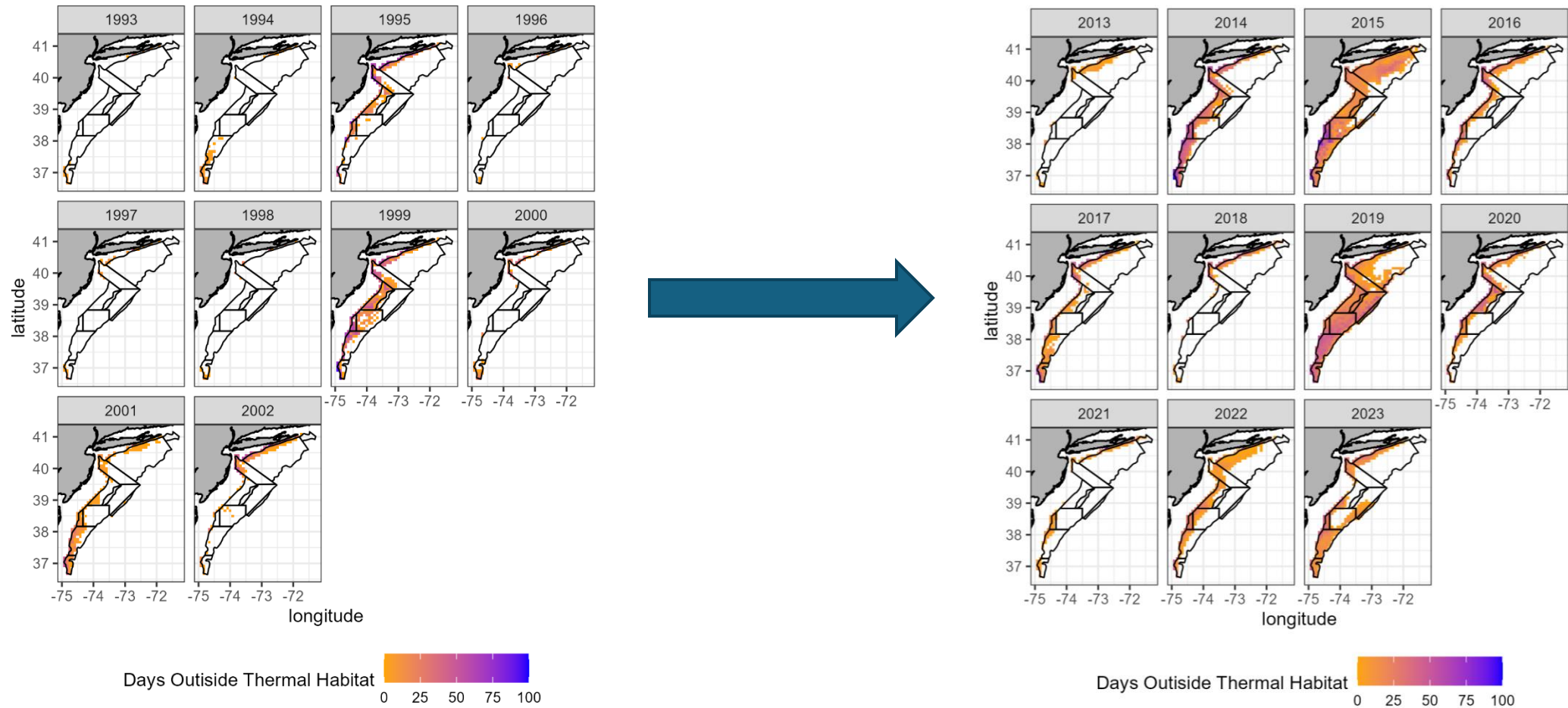
Characterizing Baseline Thermal Habitat



Baseline thermal habitat defined as areas where maximum monthly bottom temperature always $< 18^{\circ}\text{C}$ between 25 and 100m

Large decrease in thermal habitat in recent decade within inshore and southern MAB.

Duration of Heat Stress Events

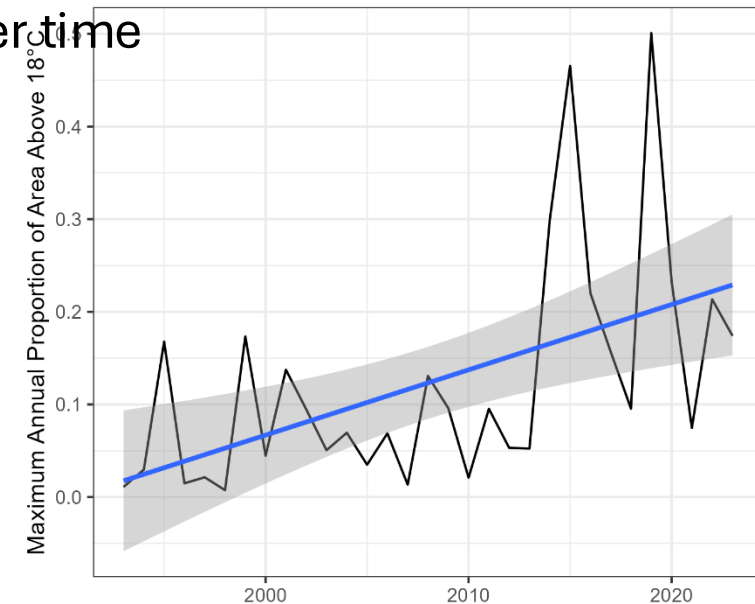
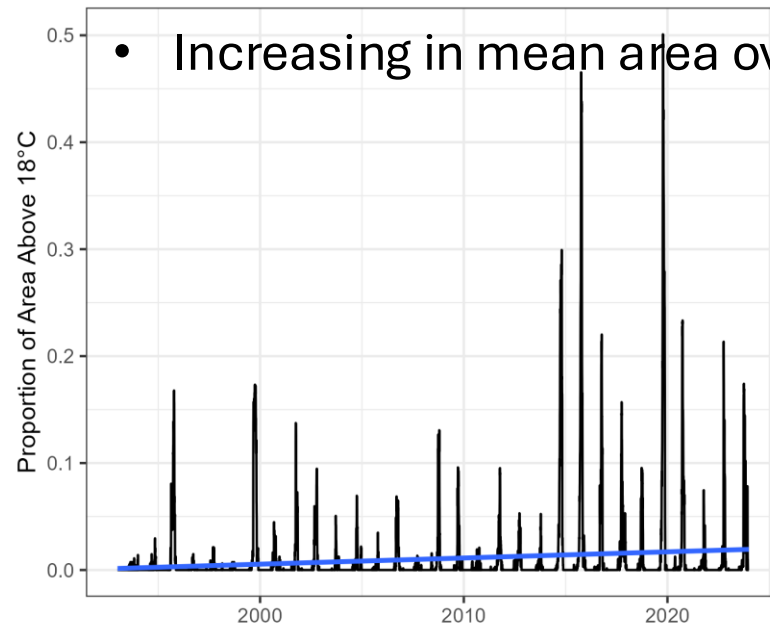
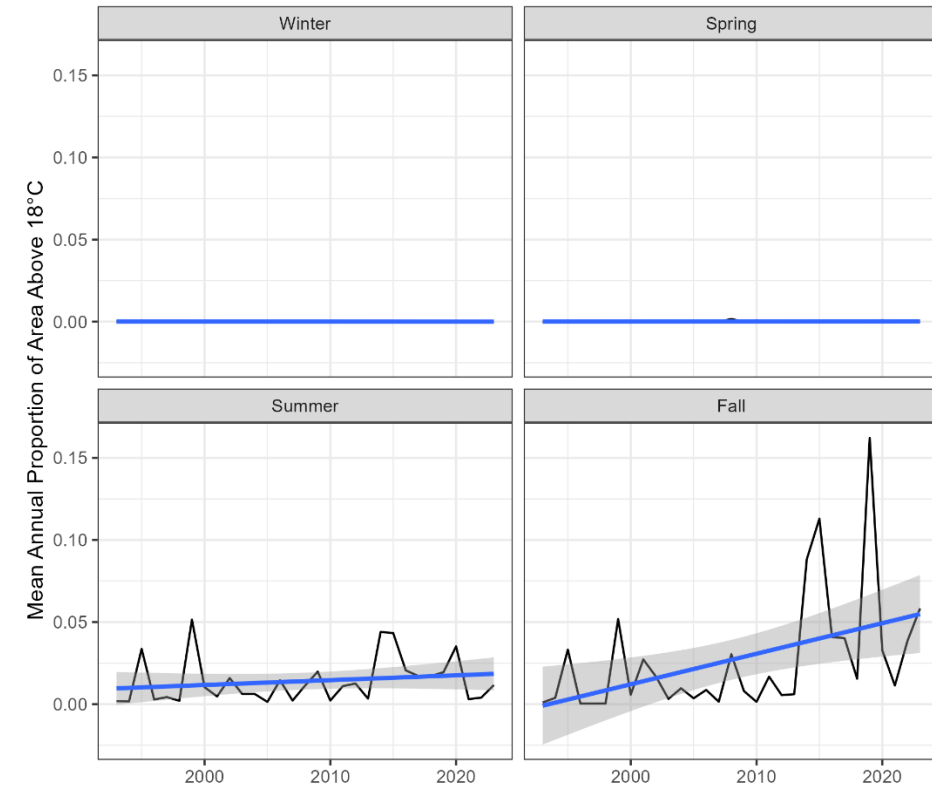


Heat stress events when bottom temperature $> 18^{\circ}\text{C}$ and can occur from days to weeks
Large increase in heat stress events in recent decade (especially 2015 and 2019)

Extent of Heat Stress Events

- Peak daily extent in last 10 years, 7-50% of managed area
- Mostly occur in summer (June-August) and Fall (Sept-Dec.)

- Increasing in mean area over time



Conclusions

1. Baseline thermal habitat for sea scallops is contracting in the MAB
2. There is an increase in the extent and duration of heat stress events, especially in September and October
3. Heat stress doesn't fully explain observed mortality events. Other possible factors include:
 - Food availability
 - Bottom oxygen
 - Scallop energy reserves (e.g., glycogen)

Ocean Acidification (OA) and Sea Scallops

Pousse et al. (2023) exposed juvenile sea scallops to varying levels of CO₂. Higher CO₂ was associated with reduced feeding rates and slower growth. Studies with other pectinids show elevated levels of larval mortality with exposure to OA (e.g., White et al 2014).

Cooley et al. (2015) and Rheuban et al. (2018) modelled the effects of OA and warming on sea scallops. They found that OA may start having measurable effects on growth and recruitment by mid-century.

While subtle effects may already be occurring, more direct effects of OA are likely in the future

Astropecten americanus

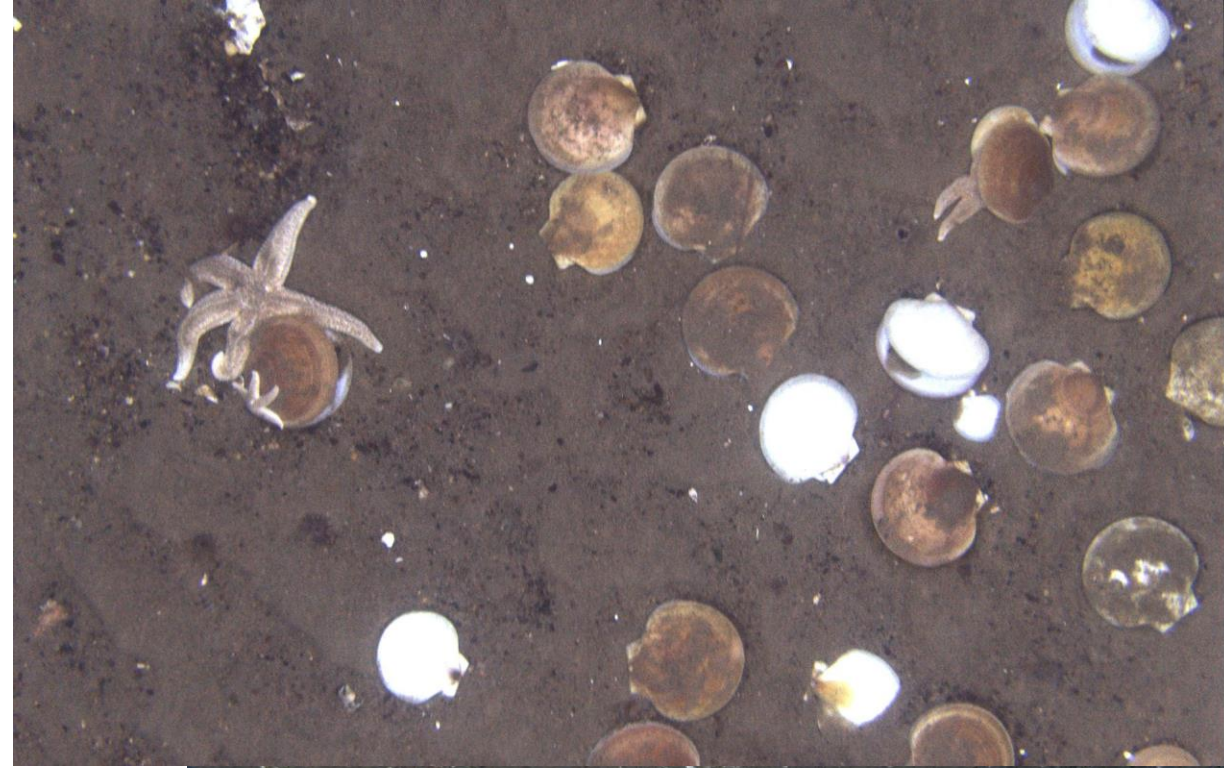
The sea star *Astropecten americanus* is common in deeper Mid-Atlantic waters. They consume small invertebrates ($< 15\text{mm}$), including juvenile sea scallops. They have been shown to reduce or exclude scallop recruitment (Hart 2006, Shank et al. 2012).

A. americanus is cold sensitive, and is rarely observed in areas where minimum bottom temperatures are less than 5 C (Franz et al. 1981). They are spreading to shallower and more northerly waters in recent years, including the southern flank of Georges Bank, and may be in part responsible for recent poor recruitment in the Mid-Atlantic



Other Major Predators

Asterias vulgaris and *A. forbesi* are classic sea star predators of shellfish, including scallops, prying them open and everting their stomachs. *A. vulgaris* is the colder water species, while *A. forbesi* prefers warmer water. There is evidence of elevated sea star predation on scallops in 2024.

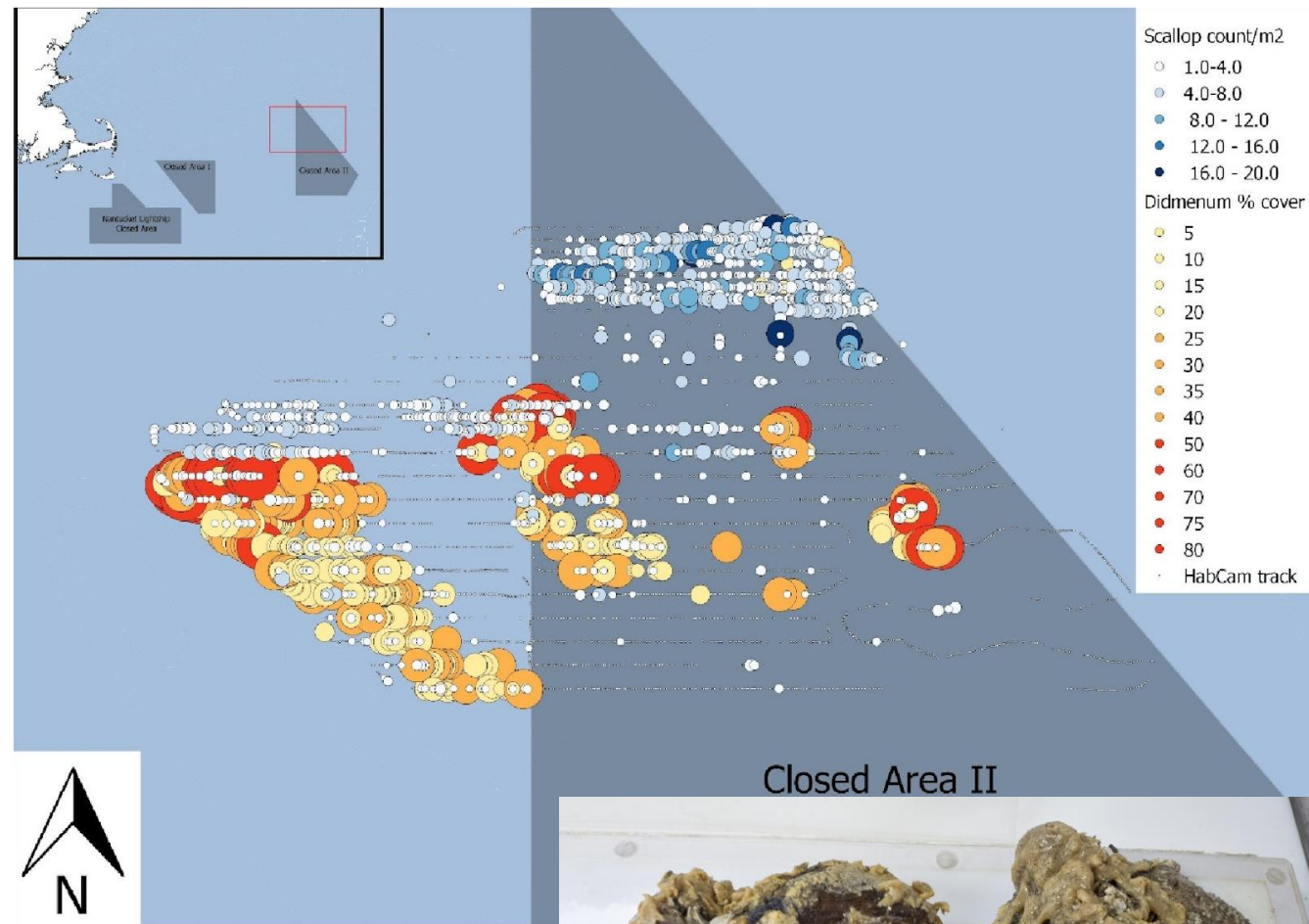
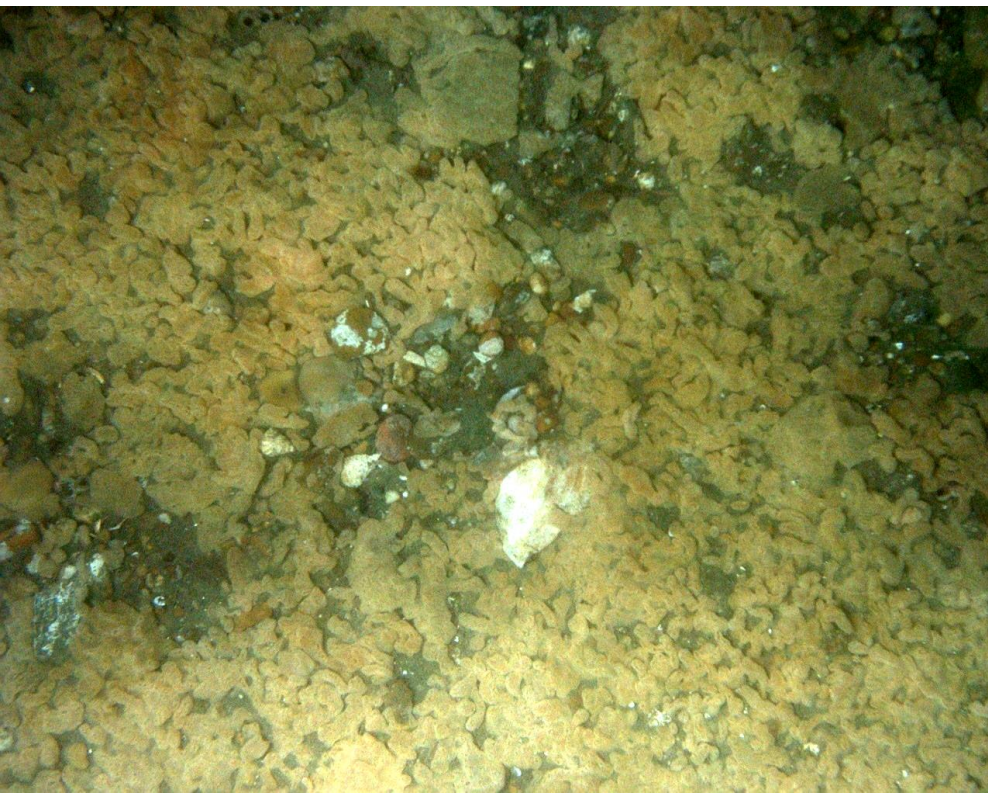


Cancer spp. crabs are major predators of juvenile (< 90mm) scallops, and are likely agents of density dependence that can induce high mortality on large year classes (Hart and Shank 2011)



Didemnum vexillum

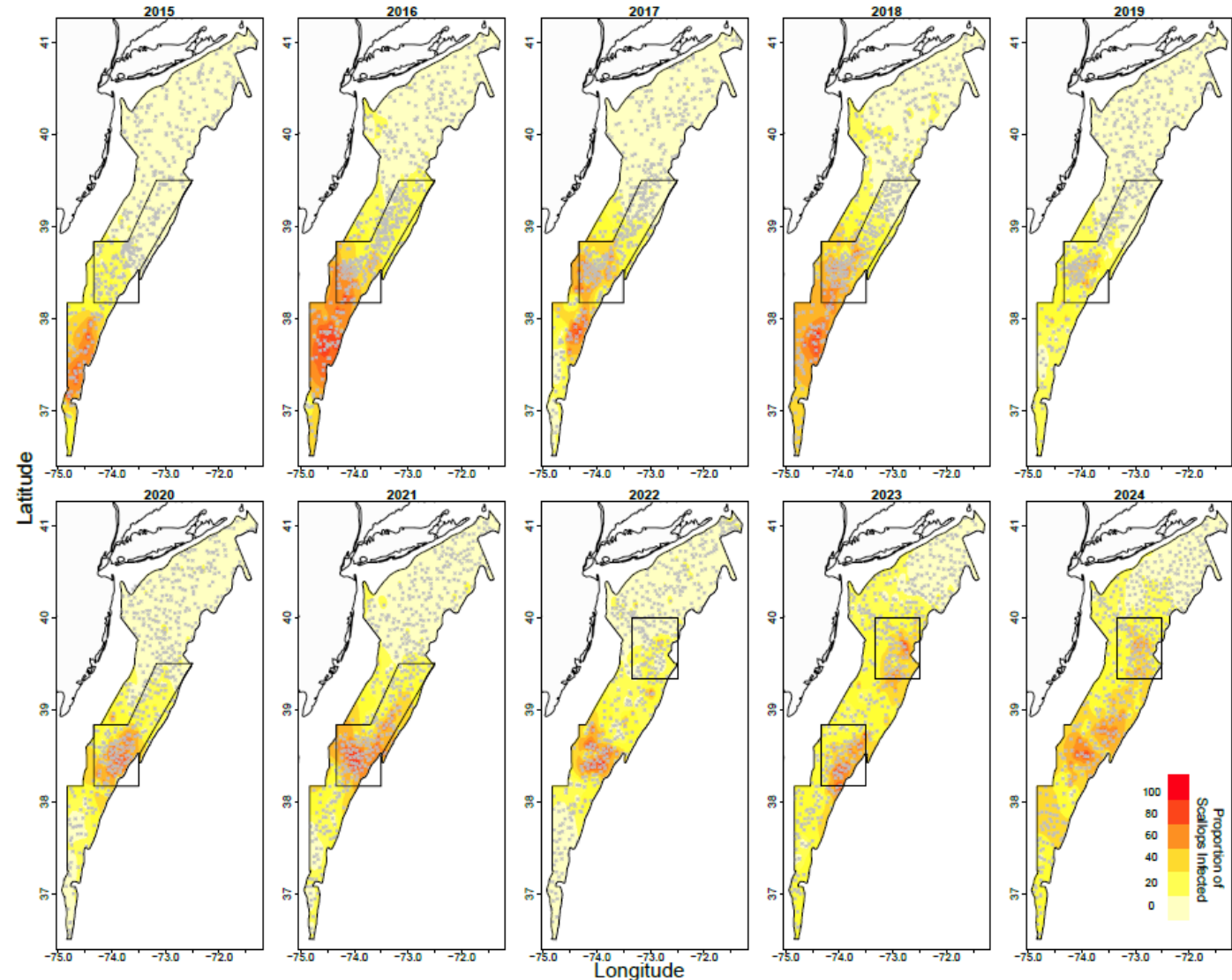
The invasive carpet tunicate *Didemnum vexillum* has become established on portions of Georges Bank, mainly the northern edge. Scallop spat cannot settle on *Didemnum*, and thus this tunicate appears to be reducing scallop recruitment (Kaplan et al 2017).



Scallop Parasites and Diseases

Nematodes

Many Mid-Atlantic scallops in recent years have been infected by a larval stage of the nematode *Sulcascaris sulcate*. The adult host of these worms are sea turtles, who acquire the parasite by consuming infected scallops. (Rudders et al 2023).



Scallop Parasites and Diseases

There have been outbreaks of “gray meat” disease on Georges Bank, most notably in 2010-12, causing elevated mortality. At least one cause has been determined to be Apicomplexa protists, although there may be other causes as well (Levesque et al. 2016, Inglis et al. 2016, Siemann et al. 2019)

Scallop “blister disease” has become an increasing problem in recent years, mostly in the Mid-Atlantic. The scallop shell get colonized underneath the gonad, and then the scallop forms a shell blister to try to protect itself. Blistered scallops can have reduced meat weight and severe cases can cause mortality.

